

Sample Mid Question

Time: 70 min

Full marks: 30

Instructions:

- The output format should be strictly maintained. Every deviation from the given output will reduce marks.
- Create a file **yourID_prob1.c** and **yourID_prob2.c** on the desktop inside a folder named **yourID_Online1_CSE282**. Code there and submit the .c files in google classroom.
- Name all your variables as **VariableName_XX**, where XX= last 2 digits of your StudentID. For example, name variable “QC_name” as “QC_name_28” for ID 202036028.

After graduating in IPE from MIST, you joined as a Production Quality Control (QC) Engineer at Samsung’s new smartphone manufacturing plant in Bangladesh. While most of their production line is automated, a final check of the phone is still done by a human, a *QC specialist*. As a QC Engineer, you decided to automate this process as far as you can. Now, automate the following scenarios using C programming.

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Each day, between 1-5 production lines are kept operational based on demand and other factors. Each production line ends with the final QC check, which is, as mentioned before, done by a single QC Specialist. He/she checks finished smartphones in batches, with **each batch containing 100 phones**.

Now, take input of how many production lines were active today, how many batches of smartphones were checked by QC the specialist in each line and how many phones passed the QC test in each batch as shown in the table below.

Then, **for each active line**, print the error percentage per batch (100-Number of phones that passes QC), total average error% and the QC Specialist’s daily Salary, which is:

Daily Salary = 100 * number of batches checked + 10 * total number of QC failed smartphones

Also, print a warning message if Total Error percentage > 5% as shown in the table below.

Input	Output
How Many QC lines were active today: 2	QC Line 1:
Enter Data for QC Line 1:	Error Percentage per batch: 3% 5% 1% 8% 2%
How Many Batches were checked: 5	Total Avg Error % today: 3.80%
Enter QC Passed in each batch: 97 95 99 92 98	Line 1 is working Fine
	Line 1 QC Salary: 690
Enter Data for QC Line 2:	QC Line 2:
How Many Batches were checked: 4	Error Percentage per batch: 4% 6% 9% 12%
Enter QC Passed in each batch: 96 94 91 88	Total Avg Error % today: 7.75%
	Warning! Line 2 is over error threshold (5%)
	Line 2 QC Salary: 710

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While making this program to automate part of the QC process, you found out that all the QC specialist’s full names are saved in your company’s database. However, suppose that for better tracking, you need the names to be separated into first and last names. Now, write a C program that takes the full names as input and prints the First and Last names of QC specialists as shown in the table below.

Consider that,

First Name = All Words Except the last word of the full name
Last Name = Last word of the full name

Print Appropriately. For example, print only the first name if the full name has only one word. Print both first and last name if the full name has more than 1 word.

Input	Output
Enter QC Specialist’s Full Name: Salman Ibrahim	First Name: Salman Last Name: Ibrahim
Enter QC Specialist’s Full Name: Tanjeel Ahmed Bin Zaman	First Name: Tanjeel Ahmed Bin Last Name: Zaman
Enter QC Specialist’s Full Name: Naurin Tabassum Shuchi	First Name: Naurin Tabassum Last Name: Shuchi
Enter QC Specialist’s Full Name: Limon	First Name: Limon

3 Print a pattern of height **n** with *, as shown in the table below. Take **n** as user input.

Input	Output
Enter the value of n: 4	**** * * * * ****
Enter the value of n: 6	***** * * * * * * * * *****